158.337
Database Development

Course Coordinator: A/P Anu Mathrani

Course Tutor: Ms. Indu Sofat



Learning Outcomes

1. Demonstrate comprehension of database theory and architecture by relating them to real world data applications.
2. Model, design and implement relational and selected non-relational databases.
3. Work with data using appropriate languages and programming tools to store, manipulate and analyze enterprise-wide data.
4. Compare and contrast various data handling, transactional and analytical processing techniques for relational and selected non-relational databases.
5. Recommend a suitable database approach for given information technology system needs.

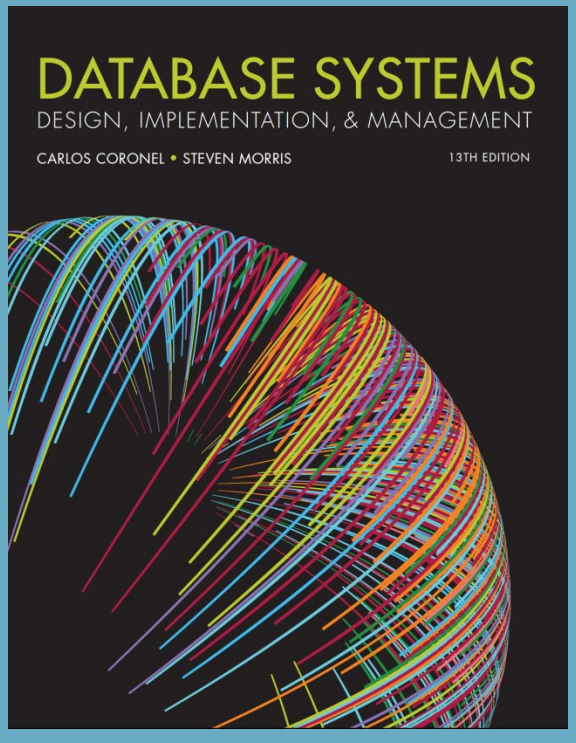
Text Book

Textbook - Database Systems: Design, Implementation and Management by Carlos Coronel and Steven Morris
Edition –13th edition
Publisher – Cengage

Other Recommended Reading

- Any Oracle textbook for getting a hands-on approach to the software platform, SQL syntax and database administration.
- http://www.oracle.com/
- https://www.mongodb.com/

Note: Lecture notes and other materials are supplied through Stream. You should read the lecture handouts before each lecture.



Chapter 1

Database Systems



Learning Objectives

- Define the difference between data and information
- Describe what a database is, various types, and why they are valuable assets for decision making
- Explain the importance of database design
- See how modern databases evolved from file systems
- Understand flaws in file system data management
- Outline the main components of the database system
- Describe the main functions of a database management system (DBMS)

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Why Databases?

- Characteristics of data in today's world
  - Ubiquitous (i.e., abundant, global, and everywhere)
  - Pervasive (i.e., unescapable, prevalent, and persistent)

- Databases make data persistent and shareable in a secure way
  - Specialized structures that allow computer-based systems to store, manage, and retrieve data very quickly

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Data versus Information

- Data consists of raw facts
  - Not yet processed to reveal meaning to the end user
  - Building blocks of information

- Information results from processing raw data to reveal meaning
  - Requires context
  - Bedrock of knowledge
  - Should be accurate, relevant, and timely

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Introducing the Database

- Shared, integrated computer structure that stores data
  - End-user data: raw facts of interest to end user
  - Metadata: data about data, through which the end-user data is integrated and managed
    - Describes data characteristics and relationships
- Database management system (DBMS)
  - Collection of programs
  - Manages the database structure
  - Controls access to data stored in the database

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

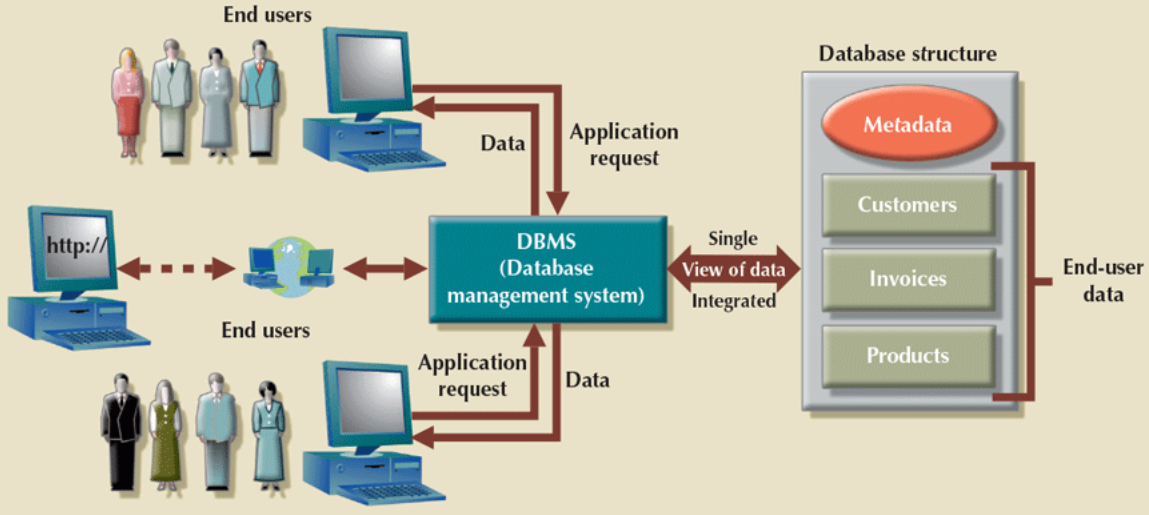
Role and Advantages of the DBMS (1 of 2)

- Database management system (DBMS): intermediary between the user and the database
  - Enables data to be shared
  - Presents the end user with an integrated view of data
  - Provides more efficient and effective data management
  - Improves sharing, security, integration, access, decision-making, productivity, etc.

CENGAGE
© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Role and Advantages of the DBMS (2 of 2)

FIGURE 1.4 THE DBMS MANAGES THE INTERACTION BETWEEN THE END USER AND THE DATABASE



© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Types of Databases (1 of 5)

- Single-user database supports one user at a time
  - Desktop database: single-user database on a personal computer
- Multiuser database: supports multiple users at the same time
  - Workgroup databases: supports a small number of users or a specific department
  - Enterprise database: supports many users across many departments

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Types of Databases (2 of 5)

- Classification by location
  - Centralized database: data located at a single site
  - Distributed database: data distributed across different sites
  - Cloud database: created and maintained using cloud data services that provide defined performance measures for the database

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Types of Databases (3 of 5)

- Classification by data type
  - General-purpose database: contains a wide variety of data used in multiple disciplines
  - Discipline-specific database: contains data focused on specific subject areas
  - Operational database: designed to support a company's day-to-day operations

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Types of Databases (4 of 5)

- Analytical database: stores historical data and business metrics used exclusively for tactical or strategic decision making
  - Data warehouse: stores data in a format optimized for decision support
  - Online analytical processing (OLAP): tools for retrieving, processing, and modeling data from the data warehouse
  - Business intelligence: captures and processes business data to generate information that support decision making

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Types of Databases (5 of 5)

- Databases can be classified to reflect the degree to which the data is structured
  - Unstructured data exists in its original (raw) state
  - Structured data results from formatting
    - Structure is applied based on type of processing to be performed
  - Semistructured data: processed to some extent

- Extensible Markup Language (XML)
  - Represents data elements in textual format

- JSON or BSON (JavaScript Object Notation or Binary JSON)
  - JSON represents a lightweight data-interchange format (for humans)
  - BSON is binary-encoded format for JSON data for computers)

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Why Database Design Is Important

- Focuses on design of database structure that will be used to store and manage end-user data
  - Well-designed database: facilitates data management and generates accurate and valuable information
  - Poorly designed database: can cause difficult-to-trace errors that may lead to poor decision making

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Evolution of File System Data Processing (1 of 3)

- Manual file systems
  - Accomplished through a system of file folders and filing cabinets
- Computerized file systems
  - Data processing (DP) specialist created a computer-based system to track data and produce required reports
- File system redux: modern end-user productivity tools
  - Includes spreadsheet programs such as Microsoft Excel

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Evolution of File System Data Processing (2 of 3)

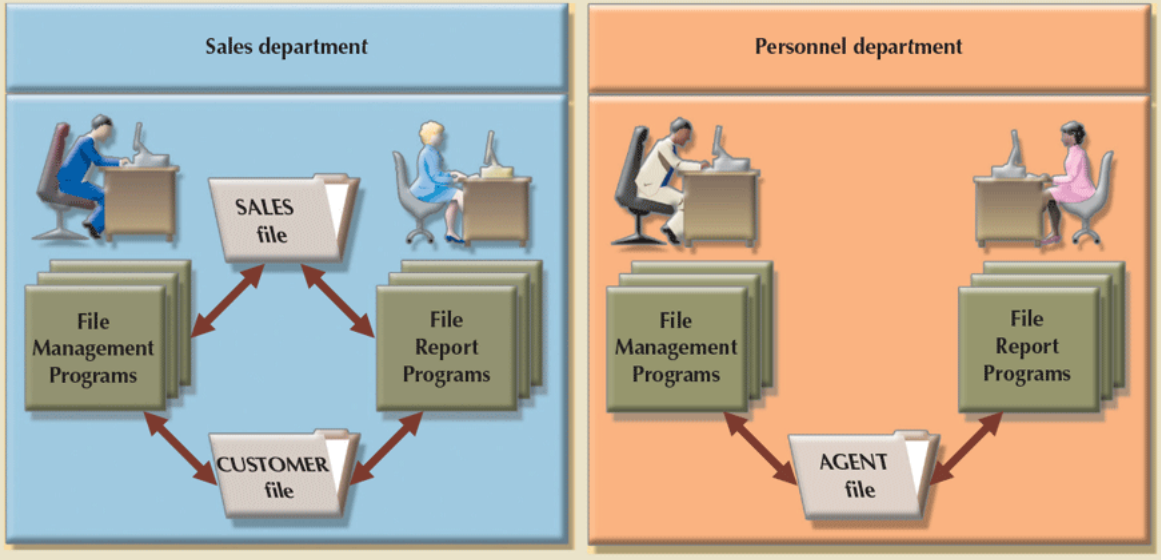
Table 1.2	Basic File Terminology
TERM	DEFINITION
Data	Raw facts, such as a telephone number, a birth date, a customer name, and a year-to-date (YTD) sales value. Data has little meaning unless it has been organized in some logical manner.
Field	A character or group of characters (alphabetic or numeric) that has a specific meaning. A field is used to define and store data.
Record	A logically connected set of one or more fields that describes a person, place, or thing. For example, the fields that constitute a record for a customer might consist of the customer's name, address, phone number, date of birth, credit limit, and unpaid balance.
File	A collection of related records. For example, a file might contain data about the students currently enrolled at Gigantic University.

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Evolution of File System Data Processing (3 of 3)

FIGURE 1.9 A SIMPLE FILE SYSTEM



© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Problems with File System Data Processing

- Problems with file systems challenge the types of information that can be created from data as well as information accuracy
  - Lengthy development times
  - Difficulty of getting quick answers
  - Complex system administration
  - Lack of security and limited data sharing
  - Extensive programming

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Structural and Data Dependence (1 of 2)

- Structural dependence
  - Access to a file is dependent on its own structure
  - All file system programs are modified to conform to a new file structure
- Structural independence
  - File structure is changed without affecting the application's ability to access the data

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Structural and Data Dependence (2 of 2)

- Data dependence
  - Data access changes when data storage characteristics change
- Data independence
  - Data storage characteristics are changed without affecting the program's ability to access the data
- Practical significance of data dependence is the difference between logical and physical format

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Data Redundancy

- Unnecessarily storing the same data at different places
  - Islands of information (i.e., scattered data locations)
  - Increases the probability of having different versions of the same data

- Possible results of uncontrolled data redundancy
  - Poor data security
  - Data inconsistency
  - Data-entry errors
  - Data integrity problems

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Data Anomalies

- Develop when not all of the required changes in the redundant data are made successfully
  - Update anomalies
  - Insertion anomalies
  - Deletion anomalies

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Database Systems (1 of 2)

- Logically related data stored in a single logical data repository
  - Physically distributed among multiple storage facilities
  - DBMS eliminates most of file system's data inconsistency, data anomaly, data dependence, and structural dependence problems

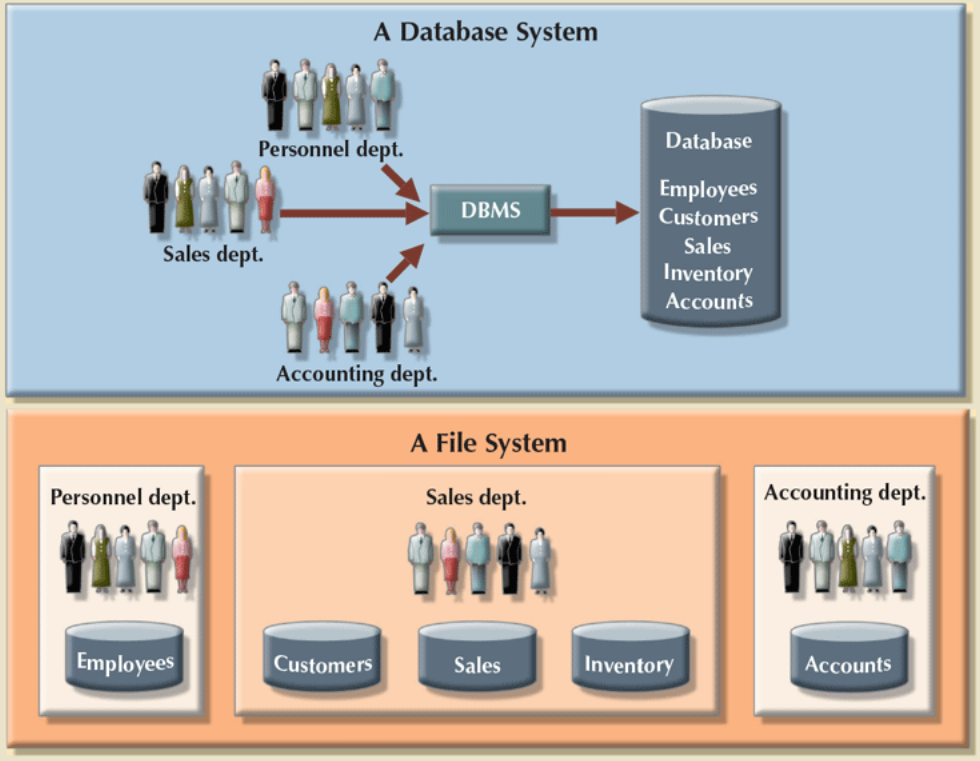
- Current generation DBMS software
  - Stores data structures, relationships between structures, and access paths
  - Defines, stores, and manages all access paths and components

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Database Systems (2 of 2)

FIGURE 1.10 CONTRASTING DATABASE AND FILE SYSTEMS

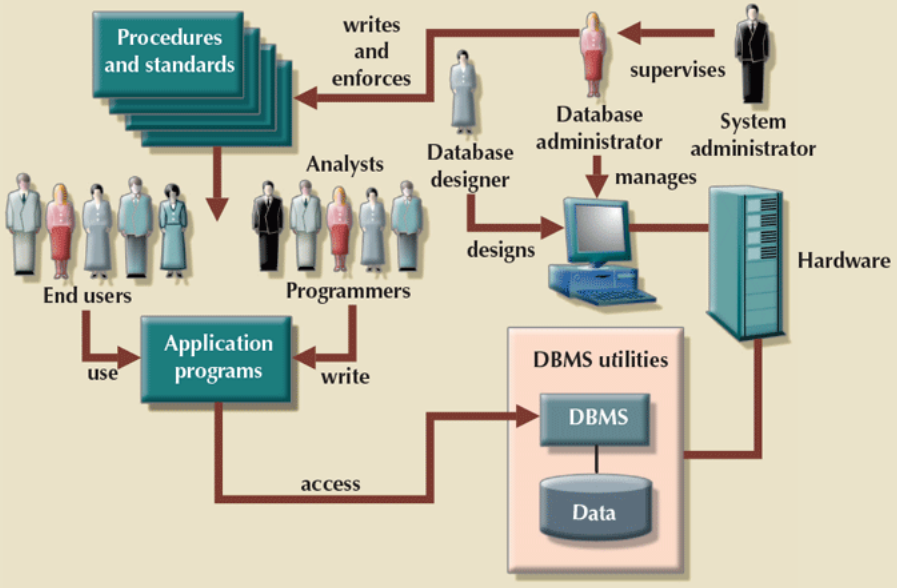


© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Database System Environment

- Database system: organization of components that define and regulate the collection, storage, management, and use of data within a database environment
  - Hardware
  - Software
  - People
  - Procedures
  - Data

FIGURE 1.11 THE DATABASE SYSTEM ENVIRONMENT



© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



DBMS Functions (1 of 3)

- Data dictionary management
  - Data dictionary: stores definitions of data elements and their relationships
- Data storage management
  - Performance tuning ensures efficient performance
- Data transformation and presentation
  - Data is formatted to conform to logical expectations
- Security management
  - Enforces user security and data privacy

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

DBMS Functions (2 of 3)

- Multiuser access control
  - Sophisticated algorithms ensure that multiple users can access the database concurrently without compromising its integrity
- Backup and recovery management
  - Enables recovery of the database after a failure
- Data integrity management
  - Minimizes redundancy and maximizes consistency

CENGAGE
© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



DBMS Functions (3 of 3)

- Database access languages and application programming interfaces
  - Query language: lets the user specify what must be done without having to specify how
  - Structured Query Language (SQL): de facto query language and data access standard supported by the majority of DBMS vendors
- Database communication interfaces
  - Accept end-user requests via multiple, different network environments

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Managing the Database System: A Shift in Focus

- Disadvantages of database systems
  - Increased costs
  - Management complexity
  - Maintaining currency
  - Vendor dependence
  - Frequent upgrade/replacement cycles

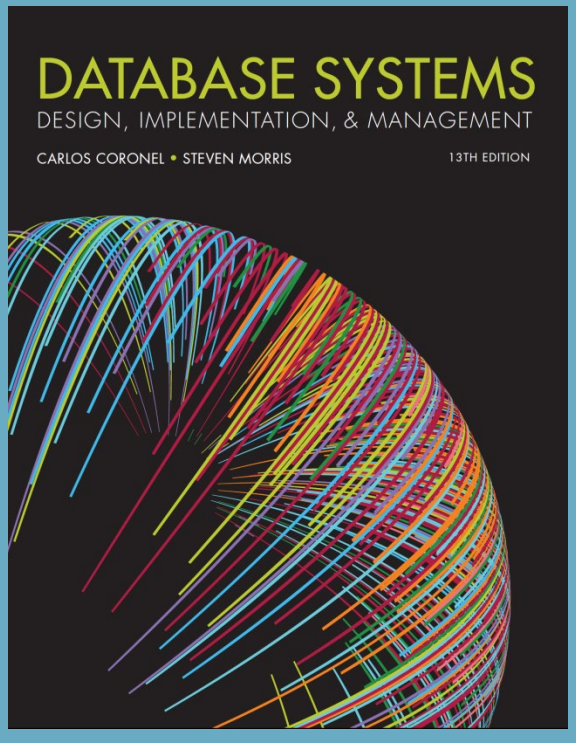
CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Preparing for Your Database Professional Career

TABLE 1.3	DATABASE CAREER OPPORTUNITIES	
JOB TITLE	DESCRIPTION	SAMPLE SKILLS REQUIRED
Database Developer	Create and maintain database-based applications	Programming, database fundamentals, SQL
Database Designer	Design and maintain databases	Systems design, database design, SQL
Database Administrator	Manage and maintain DBMS and databases	Database fundamentals, SQL, vendor courses
Database Analyst	Develop databases for decision support reporting	QL, query optimization, data warehouses
Database Architect	Design and implementation of database environments (conceptual, logical, and physical)	DBMS fundamentals, data modeling, SQL, hardware knowledge, etc.
Database Consultant	Help companies leverage database technologies to improve business processes and achieve specific goals	Database fundamentals, data modeling, database design, SQL, DBMS, hardware, vendor-specific technologies, etc.
Database Security Officer	Implement security policies for data administration	DBMS fundamentals, database administration, SQL, data security technologies, etc.
Cloud Computing Data Architect	Design and implement the infrastructure for next-generation cloud database systems	Internet technologies, cloud storage technologies, data security, performance tuning, large databases, etc.
Data Scientist	Analyze large amounts of varied data to generate insights, relationships, and predictable behaviors	Data analysis, statistics, advanced mathematics, SQL, programming, data mining, machine learning, data visualization

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Chapter 2

Data Models



Learning Objectives

- Discuss data modeling and why data models are important
- Describe the basic data-modeling building blocks
- Define what business rules are and how they influence database design
- Understand how the major data models evolved
- List emerging alternative data models and the needs they fulfill
- Explain how data models can be classified by their level of abstraction

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Data Modeling and Data Models

- Data modeling: creating a specific data model for a determined problem domain
  - Data model: simple representation of complex real-world data structures
    - Useful for supporting a specific problem domain
  - Model: abstraction of a more complex real-world object or event

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



The Importance of Data Models

- The importance of data modeling cannot be overstated
  - Facilitates communication
  - Gives various views of the database
  - Organizes data for various users
  - Provides an abstraction for the creation of good a database

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Data Model Basic Building Blocks

- Entity: person, place, thing, or event about which data will be collected and stored
  - Attribute: characteristic of an entity
  - Relationship: association among entities
    - One-to-many (1:M OR 1..*)
    - Many-to-many (M:N or *..*)
    - One-to-one (1:1 OR 1..1)
  - Constraint: restriction placed on data
    - Ensures data integrity

CENGAGE
© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Business Rules

- Brief, precise, and unambiguous description of a policy, procedure, or principle
  - Create and enforce actions within that organization's environment
  - Establish entities, relationships, and constraints

- Sources of business rules
  - Company managers
  - Policy makers
  - Department managers
  - Written documentation
  - Direct interviews with end users

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Basic Rules for Modelling Entities

- Each entity should have multiple occurrences
  - Entity instance (or occurrence) is row in table
  - Entity set is collection of like entities

- Each entity must have at least one descriptive attribute (e.g., Student_Name, Student_DOB). The number of attributes refers to the degree of the entity.

- Entities must not be free-standing (i.e., must participate in at least one relationship).

- Entities must have some unique identifier that helps to identify a particular occurrence of an entity (or primary key)

Basic Rules for Modelling Relationships

- **Participants** are entities that participate in a relationship
- Relationships between entities always operate in both directions

- **Connectivity** - Describes the relationship classification

- **Cardinality** - Expresses minimum and maximum number of entity occurrences associated with one occurrence of related entity

- **Relationships** are established by very concise statements known as business rules



Discovering Business Rules

- Reasons for identifying and documenting business rules
  - Standardize company's view of data
  - Facilitate communications tool between users and designers
  - Assist designers
    - Understand the nature, role, scope of data, and business processes
    - Develop appropriate relationship participation rules and constraints
    - Create an accurate data model

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Translating Business Rules into Data Model Components

- Business rules set the stage for the proper identification of entities, attributes, relationships, and constraints
  - Nouns translate into entities
  - Verbs translate into relationships among entities

- Relationships are bidirectional
  - Questions to identify the relationship type
    - How many instances of B are related to one instance of A?
    - How many instances of A are related to one instance of B?

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Naming Conventions

- Entity name requirements
  - Be descriptive of the objects in the business environment
  - Use terminology that is familiar to the users
- Attribute name
  - Required to be descriptive of the data represented by the attribute
- Proper naming
  - Facilitates communication between parties
  - Promotes self-documentation

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Hierarchical and Network Models (1 of 2)

- **Hierarchical models:** developed to manage large amounts of data for complex manufacturing projects
  - Represented by an upside-down tree which contains segments
    - Segments are the equivalent of a file system's record type
  - Depicts a set of one-to-many (1:M) relationships

- **Network models:** created to represent complex data relationships effectively
  - Improved database performance and imposed a database standard
  - Allows a record to have more than one parent (M:N relationship)

CENGAGE
© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Hierarchical and Network Models (2 of 2)

- Standard database concepts that emerged with the network model are still used by modern data models
  - Schema: Conceptual organization of the entire database as viewed by the database administrator
  - Subschema: Portion of the database seen by the application programs that produce the desired information from the data within the database
  - Data manipulation language (DML): Environment in which data can be managed and is used to work with the data in the database
  - Data definition language (DDL): Enables the database administrator to define the schema components

CENGAGE
© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



The Relational Model (1 of 3)

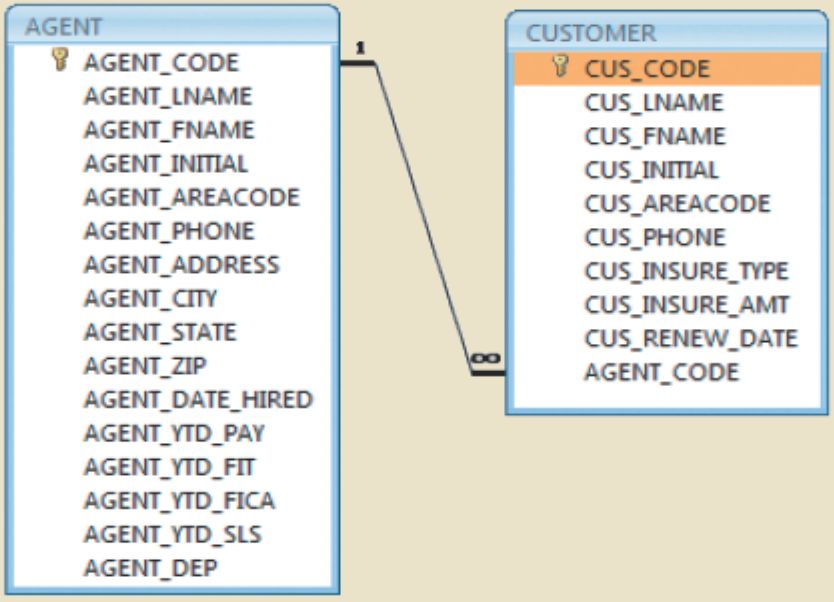
- Produced an automatic transmission database that replaced standard transmission databases
  - Based on a relation (i.e., table): matrix composed of intersecting tuples (rows) and attributes (columns)
- Describes a precise set of data manipulation constructs

- Relational database management system (RDBMS)
  - Performs basic functions provided by the hierarchical and network DBMS systems
  - Makes the relational data model easier to understand and implement
  - Hides the complexities of the relational model from the user
  - Adopts a declarative approach to data processing, i.e., specifies *what* is to be retrieved rather than *how* it can be retrieved.

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Relational Model (2 of 3)

FIGURE 2.2 A RELATIONAL DIAGRAM





© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Relational Model (3 of 3)

- SQL-based relational database application
  - End-user interface
    - Allows end user to interact with the data
  - Collection of tables stored in the database
    - Each table is independent from another
    - Rows in different tables are related based on common values in common attributes
  - SQL engine
    - Executes all queries

```
SELECT  *
FROM    CUSTOMER
WHERE   CUS_LNAME = "Smith";
```

DATA MODEL	DATA INDEPENDENCE	STRUCTURAL INDEPENDENCE
File system	No	No
Hierarchical	Yes	No
Network	Yes	No
Relational	Yes	Yes
Object Oriented	Yes	Yes

CENGAGE
© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

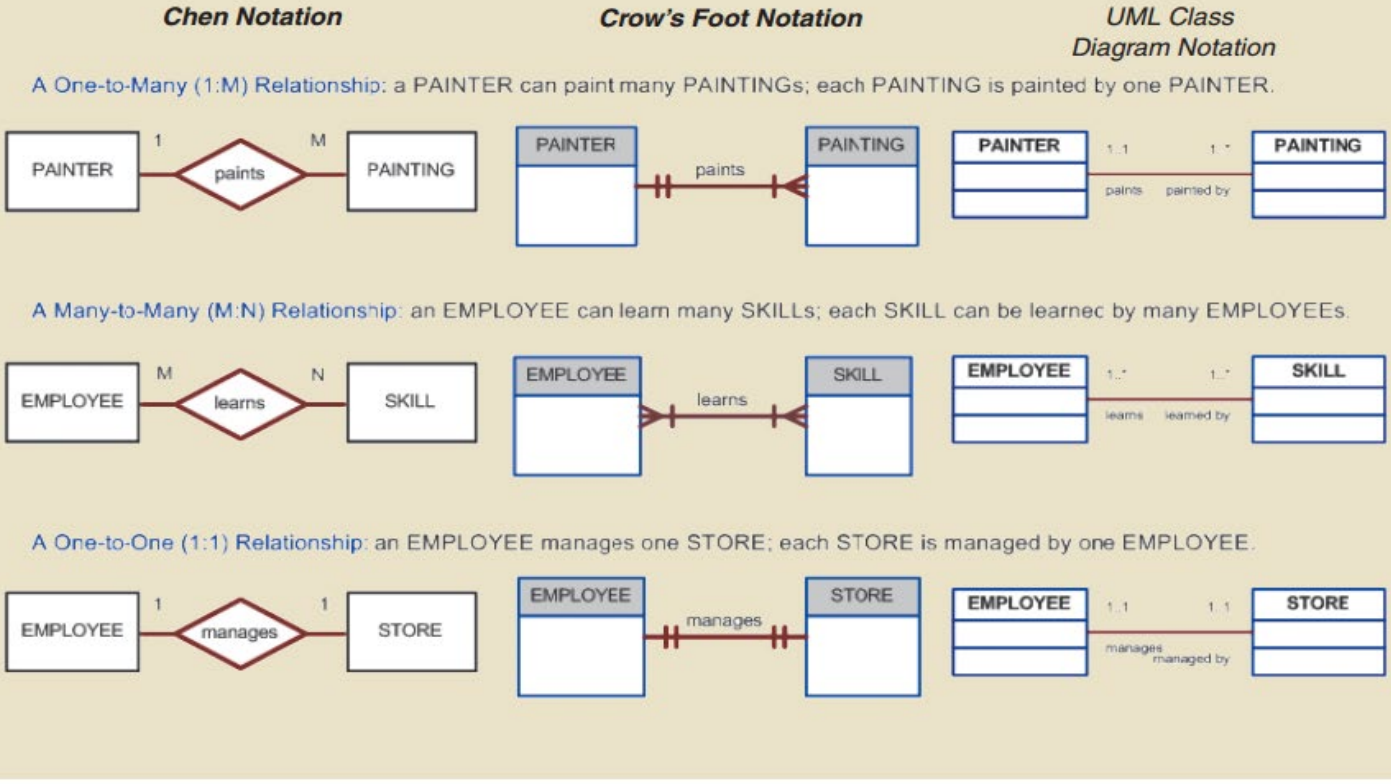
The Entity Relationship Model (1 of 2)

- Graphical representation of entities and their relationships in a database structure
  - Entity relationship diagram (ERD): uses graphic representations to model database components
  - Entity instance or entity occurrence: rows in the relational table
  - Attributes: describe characteristics
  - Connectivity: term used to label the relationship types

CENGAGE
© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Entity Relationship Model (2 of 2)

FIGURE 2.3 THE ER MODEL NOTATIONS



© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



The Object-Oriented Data Model (1 of 3)

- Both data and its relationships are contained in a single structure known as an object
  - Object-oriented database management system(OODBMS): based on OODM
- Object: contains data and their relationships with operations that are performed on it
  - Basic building block for autonomous structures
  - Abstraction of real-world entity
- Attribute: describes the properties of an object

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



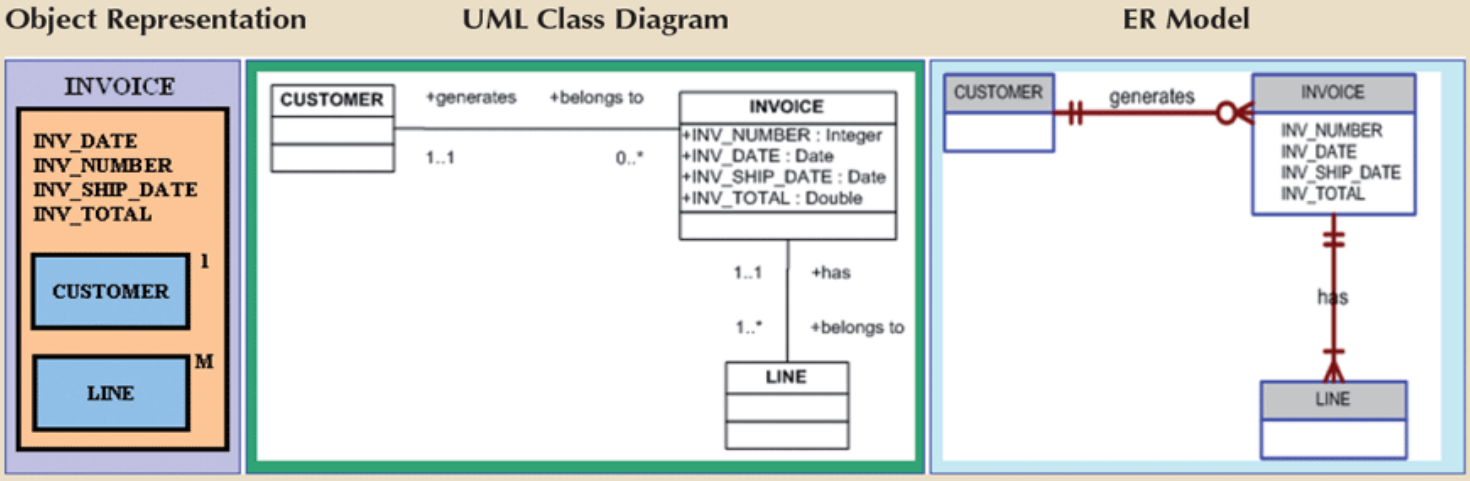
The Object-Oriented Data Model (2 of 3)

- Class: collection of similar objects with shared structure and behavior organized in a class hierarchy
- Class hierarchy: resembles an upside-down tree in which each class has only one parent
- Inheritance: object inherits methods and attributes of classes above it
- Unified Modeling Language (UML): describes sets of diagrams and symbols to graphically model a system

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Object-Oriented Data Model (3 of 3)

FIGURE 2.4 A COMPARISON OF THE OO, UML, AND ER MODELS



© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Object/Relational and XML

- Extended relational data model (ERDM)
  - Supports OO features, extensible data types based on classes, and inheritance
    - Object/relational database management system (O/R DBMS): based on ERDM
- Extensible Markup Language (XML)
  - Manages unstructured data for efficient and effective exchange of structured, semistructured, and unstructured data

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Emerging Data Models: Big Data and NoSQL (1 of 3)

- Goals of Big Data
  - Find new and better ways to manage large amounts of web and sensor-generated data
  - Provide high performance at a reasonable cost

- Characteristics of Big Data
  - Volume
  - Velocity
  - Variety

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Emerging Data Models: Big Data and NoSQL (2 of 3)

- Challenges of Big Data
  - Volume doesn't allow usage of conventional structures
  - Expensive
  - OLAP tools proved inconsistent dealing with unstructured data

- New technologies of Big Data
  - Hadoop
  - Hadoop Distributed File System (HDFS)
  - MapReduce
  - NoSQL

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

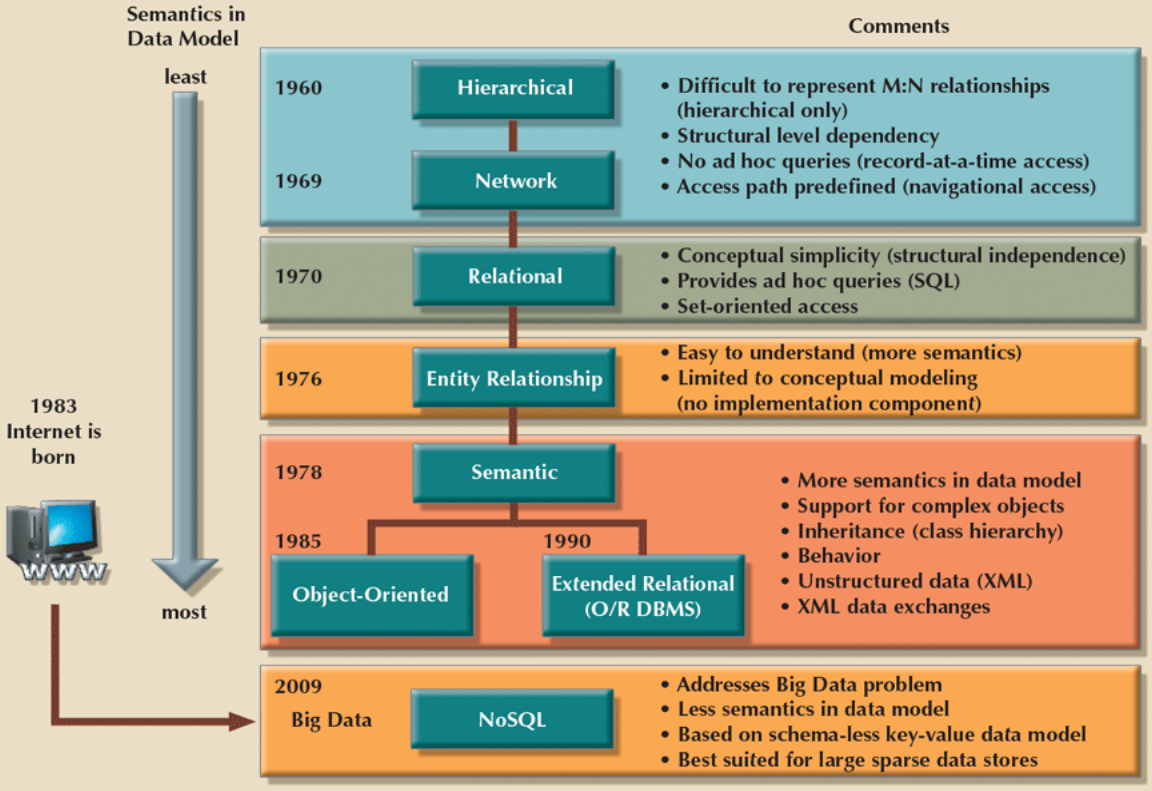
Emerging Data Models: Big Data and NoSQL (3 of 3)

- NoSQL databases
  - Not based on the relational model
  - Support distributed database architectures
  - Provide high scalability, high availability, and fault tolerance
  - Support large amounts of sparse data
  - Geared toward performance rather than transaction consistency
  - Provides a broad umbrella for data storage and manipulation

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Data Models: A Summary

FIGURE 2.5 THE EVOLUTION OF DATA MODELS



© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Hierarchical Model

- Advantages
  - Promotes data sharing
  - Parent/child relationship promotes conceptual simplicity and data integrity
  - Database security is provided and enforced by DBMS
  - Efficient with 1:M relationships
- Disadvantages
  - Requires knowledge of physical data storage characteristics
  - Navigational system requires knowledge of hierarchical path
  - Changes in structure require changes in all application programs
  - Implementation limitations
  - No data definition
  - Lack of standards

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Network Model

- Advantages
  - Conceptual simplicity
  - Handles more relationship types
  - Data access is flexible
  - Data owner/member relationship promotes data integrity
  - Conformance to standards
  - Includes data definition language (DDL) and data manipulation language (DML)
- Disadvantages
  - System complexity limits efficiency
  - Navigational system yields complex implementation, application development, and management
  - Structural changes require changes in all application programs

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Relational Model

- Advantages
  - Structural independence is promoted using independent tables
  - Tabular view improves conceptual simplicity
  - Ad hoc query capability is based on SQL
  - Isolates the end user from physical-level details
  - Improves implementation and management simplicity
- Disadvantages
  - Requires substantial hardware and system software overhead
  - Conceptual simplicity gives untrained people the tools to use a good system poorly
  - May promote information problems

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Entity Relationship Model

- Advantages
  - Visual modeling yields conceptual simplicity
  - Visual representation makes it an effective communication tool
  - Is integrated with the dominant relational model
- Disadvantages
  - Limited constraint representation
  - Limited relationship representation
  - No data manipulation language
  - Loss of information content occurs when attributes are removed from entities to avoid crowded displays

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Object-Oriented Model

- Advantages
  - Semantic content is added
  - Visual representation includes semantic content
  - Inheritance promotes data integrity
- Disadvantages
  - Slow development of standards caused vendors to supply their own enhancements
  - Complex navigational system
  - Learning curve is steep
  - High system overhead slows transactions

CENGAGE
© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

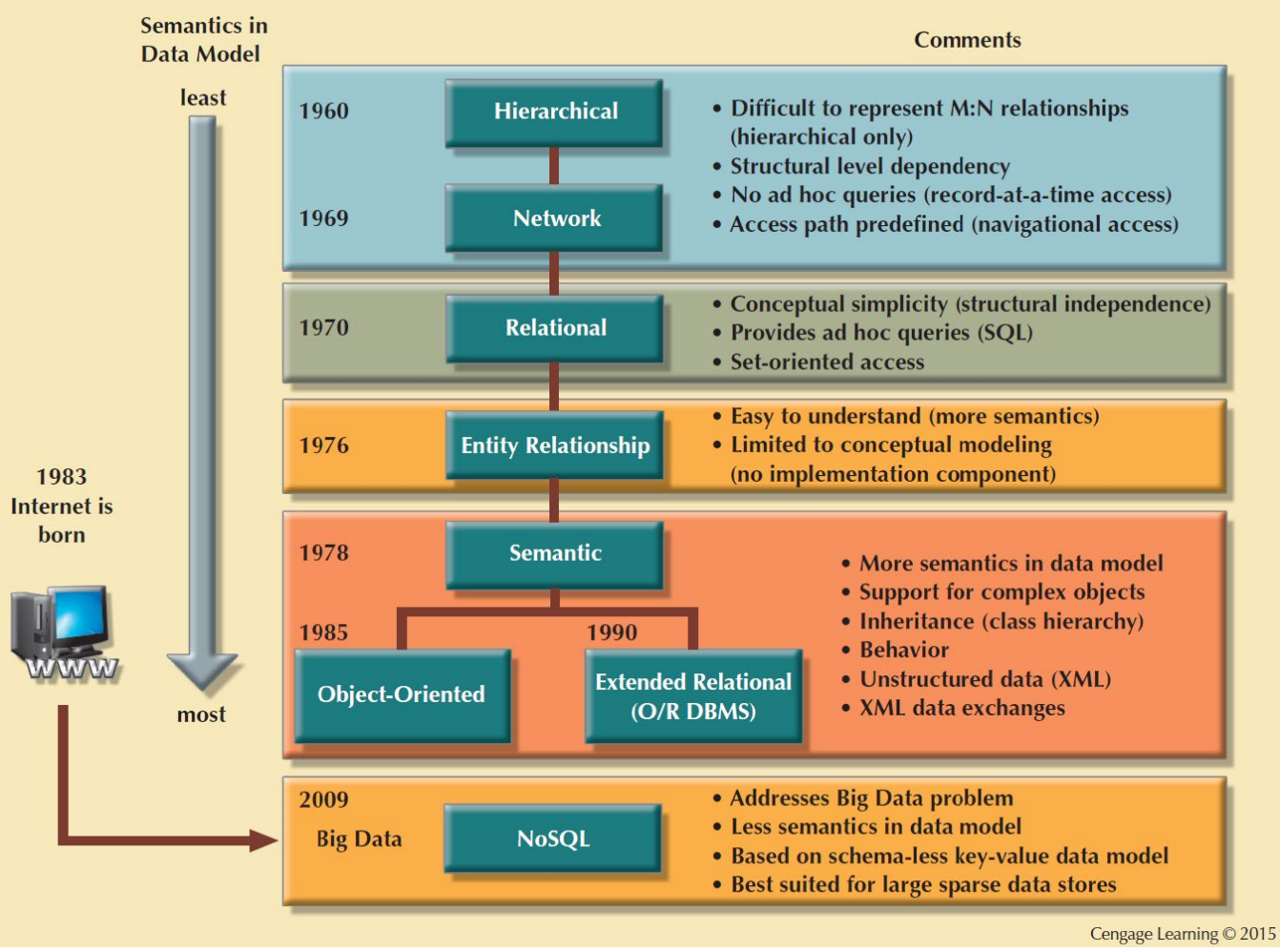


NoSQL

- Advantages
  - High scalability, availability, and fault tolerance are provided
  - Uses low-cost commodity hardware
  - Supports Big Data
  - Key-value model improves storage efficiency
- Disadvantages
  - Complex programming is required
  - There is no relationship support
  - There is no transaction integrity support
  - In terms of data consistency, it provides an eventually consistent model

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

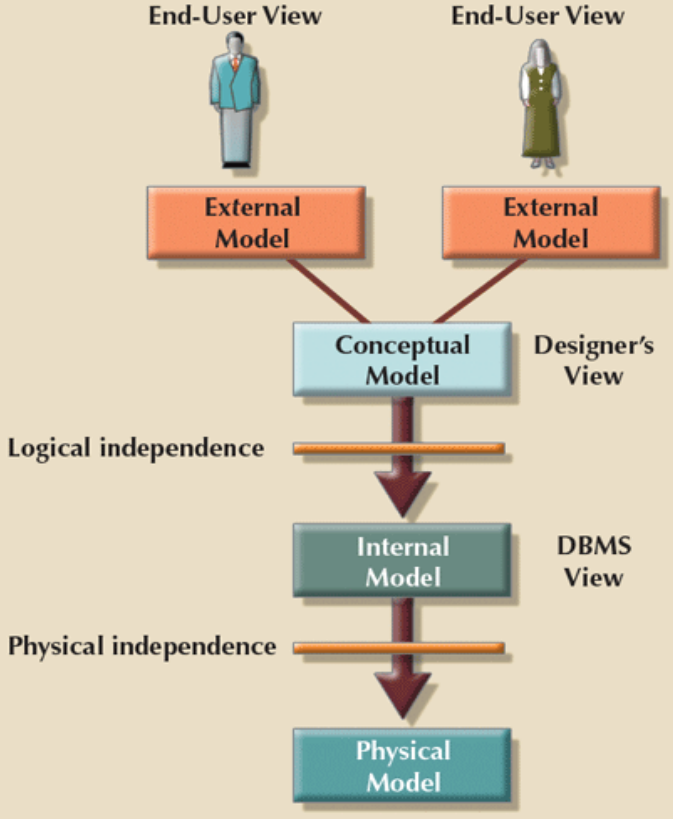
Evolution of Data Models



© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Degrees of Data Abstraction

FIGURE 2.6 DATA ABSTRACTION LEVELS



Degree of Abstraction		Characteristics
High	ER Object-Oriented	Hardware-independent Software-independent
Medium	Relational	Hardware-independent Software-dependent
Low	Network Hierarchical	Hardware-dependent Software-dependent

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



The External Model (1 of 2)

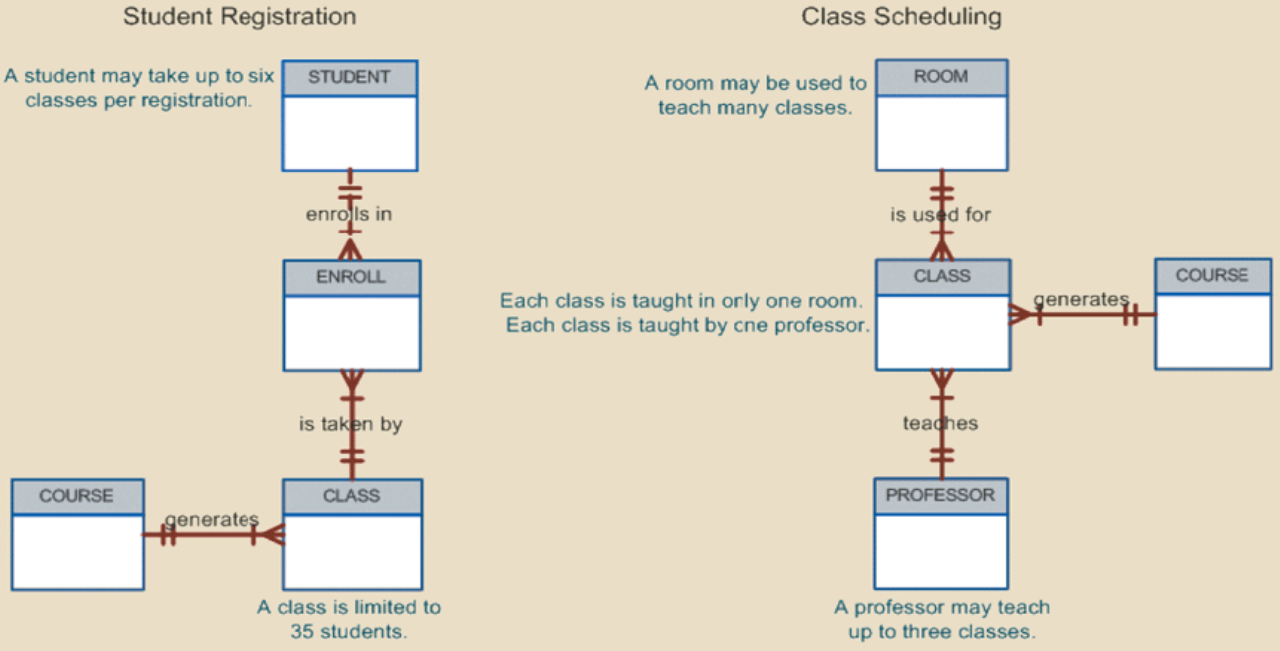
- End users' view of the data environment
  - People who use the application programs to manipulate the data and generate information

- ER diagrams are used to represent the external views
  - External schema: specific representation of an external view

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The External Model (2 of 2)

FIGURE 2.7 EXTERNAL MODELS FOR TINY COLLEGE



© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



The Conceptual Model (1 of 2)

- Represents a global view of the entire database by the entire organization
  - Conceptual schema: basis for the identification and high-level description of the main data objects
  - Logical design: task of creating a conceptual data model

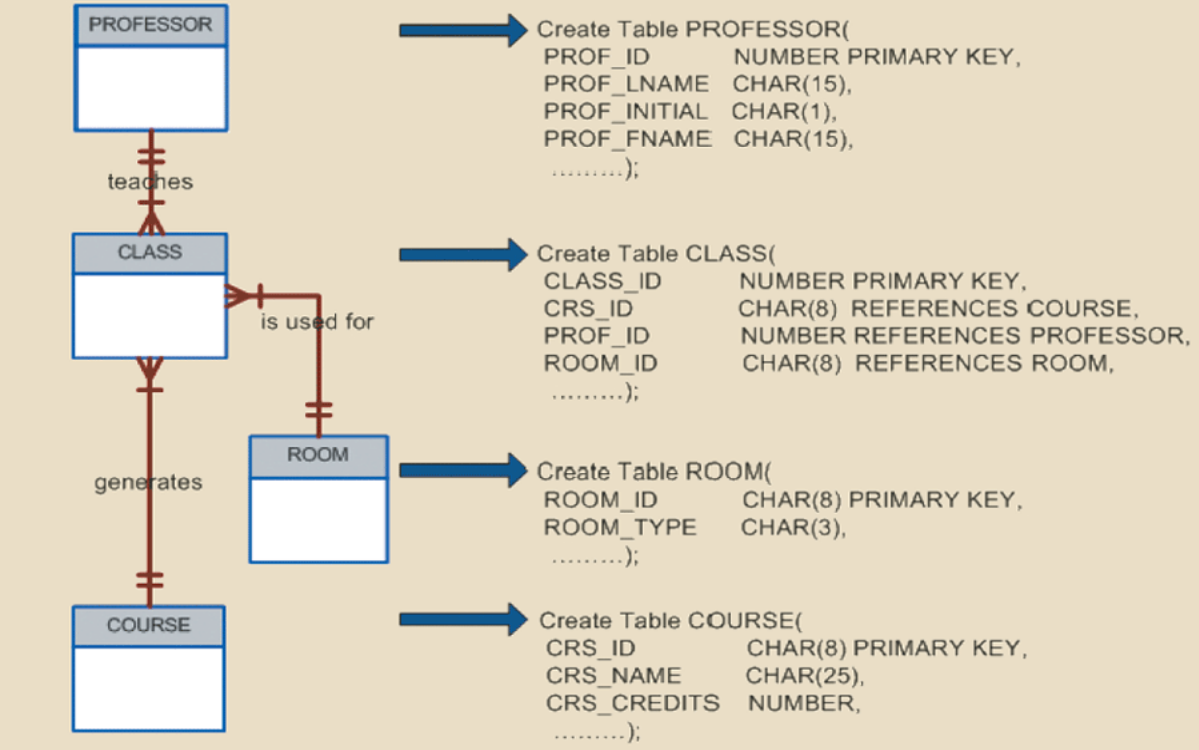
- Conceptual model advantages
  - Macro-level view of data environment
  - Software and hardware independent

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Conceptual Model (2 of 2)

FIGURE 2.9 INTERNAL MODEL FOR TINY COLLEGE



© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



The Internal Model

- Representing database as seen by the DBMS mapping conceptual model to the DBMS
  - Internal schema: specific representation of an internal model, using the database constructs supported by the chosen database
  - Logical independence: changing internal model without affecting the conceptual model
  - Hardware independent: unaffected by the type of computer on which the software is installed

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



The Physical Model

- Operates at lowest level of abstraction
  - Describes the way data are saved on storage media such as magnetic, solid state, or optical media
- Requires the definition of physical storage and data access methods
  - Software and hardware dependent
- Relational model aimed at logical level
  - Does not require physical-level details
- Physical independence: changes in physical model do not affect internal model

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Data Abstraction

Table 2.4 - Levels of Data Abstraction			
Model	Degree of Abstraction	Focus	Independent of
External	High	End-user views	Hardware and software
Conceptual	Medium-High	Global view of data (database model independent)	Hardware and software
Internal	Medium-Low	Specific database model	Hardware
Physical	Low	Storage and access methods	Neither hardware nor software

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.